

Generation Interconnection System Impact Study Report

For Queue Project Q19-1138

E Belleville 100 kV

May 20, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1138. The Interconnection Customer proposes a 199.4 MW wind facility interconnecting at the E Belleville 100 kV substation in the Illinois area. The request was received on January 17, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 397 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,260,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: E Belleville / Goodman Energy Ctr 100 KV
Contingency: E Belleville / Darlington County 100 KV
Loading: 208.9% (Rating: 136 MVA)
DFAX: 100.0%
Pre-project loading: 62.6%
2. Facility: EBelleville / Darlington County 100 KV line
Contingency: E Belleville / Goodman Energy Ctr 100 KV L
Loading: 150.8% (Rating: 189 MVA)
DFAX: 100.0%
Pre-project loading: 45.2%
3. Facility: Darlington County Substation / Ceres 100 KV
Contingency: E Belleville Substation / Goodman Energy Center 100 KV
Loading: 141.6% (Rating: 150 MVA)
DFAX: 67.5%
Pre-project loading: 51.6%
4. Facility: Alsip Substation / Goodman Energy Ctr 100 KV
Contingency: Darlington County / E Belleville 100 KV Line
Loading: 110.1% (Rating: 115 MVA)
DFAX: 43.2%
Pre-project loading: 35.1%
5. Facility: Darlington County Substation / Crystal Vly 100 KV
Contingency: E Belleville / Darlington County 100 KV
Loading: 109.6% (Rating: 123 MVA)
DFAX: 23.5%
Pre-project loading: 71.7%
6. Facility: Alsip Substation / Crystal Vly 100 KV
Contingency: Darlington County / E Belleville 100 KV
Loading: 102.3% (Rating: 171 MVA)

DFAX: 23.5%
Pre-project loading: 75.1%

7. Facility: Ceres / Opal Cliffs 100 KV
Contingency: Saticoy / Yabucoa 100 KV
Loading: 100.1% (Rating: 118 MVA)
DFAX: 23.7%
Pre-project loading: 60.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Belleville / Goodman Energy Ctr 100 KV	Rebuild line with higher-capacity conductor	\$2,788,000	60% (\$1,665,000)
EBelleville / Darlington County 100 KV line	Rebuild line with higher-capacity conductor	\$2,164,000	60% (\$1,292,000)
Darlington County Substation / Ceres 100 KV	Rebuild line with higher-capacity conductor	\$1,972,000	42% (\$832,000)
Goodman Energy Center / Alsip 100 KV	Rebuild line with higher-capacity conductor	\$1,959,000	49% (\$959,000)
Darlington County Substation / Crystal Vly 100 KV	Reconductor line	\$295,000	27% (\$80,000)
Alsip / Crystal Valley 100 KV	Reconductor line	\$551,000	27% (\$150,000)
Ceres / Opal Cliffs 100 KV	Reconductor line	\$1,005,000	28% (\$282,000)
	Total allocated to project		\$5,260,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.